

# ChunkSmith: A Scalable, Asynchronous Multimodal Retrieval-Augmented Generation System for Semantic Visual Context Preservation in Enterprise Document Analysis

<sup>1st</sup> Anshul Parate  
Department of Computer Science and  
Engineering (AIML)  
Ramdeobaba University  
Nagpur, India  
anshulnparateweb@gmail.com

<sup>2nd</sup> Ameya K. Saonerkar  
Department of Electrical  
Engineering  
Ramdeobaba University  
Nagpur, India  
saonerkarak@rknc.edu

<sup>3rd</sup> Atharva Tembhurnikar  
Department of Computer Science and  
Engineering (AIML)  
Ramdeobaba University  
Nagpur, India  
tatharva0906@gmail.com

**Abstract**—The paradigm of Retrieval-Augmented Generation (RAG) has fundamentally shifted the capabilities of Large Language Models (LLMs), allowing them to answer queries grounded in private, external data. However, a significant “Modality Gap” persists in enterprise definitions of RAG. Realworld documents—ranging from financial ledgers and scientific manuscripts to engineering schematics—encode a substantial portion of their specific knowledge in visual formats (charts, tables, diagrams). Traditional RAG pipelines, which predominantly rely on OCR-based text matching, strip away this visual context, reducing rich multimodal documents to flattened strings. This leads to hallucinations when models attempt to reason about visual trends or comparative data in tables. To address this critical deficiency, we present ChunkSmith, a productiongrade Multimodal RAG system engineered for high-fidelity visual context preservation. ChunkSmith introduces a novel SchemaFirst content processing architecture that treats images and tables not as auxiliary metadata, but as distinct, first-class semantic entities. By leveraging state-of-the-art Multimodal LLMs and a purposely designed Structured Semantic Enrichment strategy, the system generates dense vector representations that encode deep visual understanding and “predicted question” affordances. We further detail the system’s distributed backend, which employs an asynchronous Round-Robin Key Rotation algorithm to effectively bypass provider rate limits, achieving a 4.6x improvement in ingestion throughput over serial baselines. Extensive qualitative and quantitative evaluations demonstrate ChunkSmith’s superior ability to handle complex visual reasoning tasks while maintaining scalable performance metrics suitable for large corpora.

**Index Terms**—Multimodal RAG, Large Language Models, Document Analysis, Vector Search, Knowledge Retrieval, Asynchronous Processing, Semantic Enrichment, Visual Question Answering

## I. INTRODUCTION

### A. The Evolution of Retrieval Systems

The quest for intelligent information retrieval has evolved through several distinct epochs. Initial systems relied on lexical matching algorithms like TF-IDF [8] and BM25 [9], which were effective for keyword search but lacked semantic understanding. These systems operated on the principle

of term frequency analysis, treating documents as bags of words without capturing contextual meaning or relationships between concepts. The advent of Transformer-based models [10] brought about Dense Passage Retrieval (DPR) [2], enabling retrieval based on conceptual similarity rather than exact keyword matches. This represented a paradigm shift: instead of matching surface-level tokens, systems could now identify semantically related passages even when they used entirely different vocabulary. The introduction of BERT [11] and subsequent models demonstrated that pre-trained language representations could capture nuanced semantic relationships, paving the way for neural information retrieval. Most recently, Retrieval-Augmented Generation (RAG) [1] has emerged as the dominant framework for extending the knowledge boundaries of LLMs. By retrieving relevant context from a vector database and injecting it into the model’s prompt, RAG mitigates the twin problems of *hallucination* (generating factually incorrect information) and *knowledge cutoff* (inability to access recent data). This architecture has proven particularly valuable in enterprise settings where proprietary data cannot be included in model training.

### B. The Modality Gap: A Critical Limitation

Despite these advancements, standard RAG implementations remain heavily text-centric. In professional settings, arguably the most valuable information is often non-textual. Consider the following scenarios: A financial analyst querying quarterly revenue trends encoded in a multi-series line chart A medical researcher examining protein structure diagrams in a biochemistry paper An engineer referencing circuit schematics in a technical datasheet A legal professional analyzing tabular evidence in a court document. In each case, the critical information exists primarily in visual form. A sudden dip in a revenue chart, a complex chemical structure in a patent, or a specific architectural layout in a blueprint—these are the details users interact with most intensely. When a standard

RAG ingestion pipeline processes a PDF, it typically employs an Optical Character Recognition (OCR) engine like Tesseract or AWS Textract to scrape text. During this process, complex figures are either: **Ignored**: The bounding box is skipped entirely, treating the image as irrelevant noise. **Flattened**: Captions are read, but the visual data itself is lost. A chart titled "Q3 Revenue Growth" provides no information about the actual growth rate. **Corrupted**: A chart is read as a jumble of unconnected numbers. OCR might extract "2020 5M 2021 7M 2022 4M" without understanding these represent a time series. We term this loss of information the **"Modality Gap."** When a user asks, "Why did sales drop in Q3?" (referring to a bar chart), the text-only retriever finds no relevant vectors, forcing the LLM to either admit ignorance or hallucinate a plausible-sounding reason based on surrounding text. This represents a fundamental failure mode that limits RAG's applicability to real-world document analysis tasks.

### C. ChunkSmith: Bridging the Gap

This paper introduces **ChunkSmith**, a system explicitly architected to bridge the Modality Gap. Unlike prior works that attempt to solve this with end-to-end visual embedding models (which are often computationally prohibitive and less precise for text), ChunkSmith adopts a hybrid approach. It uses Multimodal LLMs as *transducers*, converting rich visual information into structured, semantically dense textual representations (summaries and potential questions) that can be indexed by standard, highly optimized vector databases. Our key insight is that while images cannot be directly embedded with the same fidelity as text, their *semantic content* can be extracted and represented textually through careful prompting of capable MLLMs. This approach combines the best of both worlds: the reasoning capabilities of large multimodal models during ingestion, and the efficiency of text-optimized retrieval during query time.

### D. Contributions

Our contributions are as follows: **Deep Semantic Extraction**: A pipeline that extracts tables and images as separate objects and uses MLLMs to generate detailed interpretations, including visual reasoning and data point extraction. **Structured Prompt Engineering**: A methodology for prompting MLLMs to act as "Question Generators," effectively pre-computing retrieval pathways and aligning document embeddings with potential user queries. **High-Throughput Architecture**: A robust asynchronous backend design with API key rotation to handle the high latency of multimodal inference, achieving near-linear speedup with the number of available API keys. **Visual Evidence Retrieval**: A user interaction model that retrieves and displays the source image alongside the AI-generated answer for verification, enhancing trust and interpretability. **Production-Ready Implementation**: A complete system built with FastAPI, React, a Vector Database, and Object Storage, demonstrating the practical viability of the approach.

## II. RELATED WORK

### A. Retrieval-Augmented Generation

The RAG paradigm was formalized by Lewis et al. [1], who demonstrated that retrieval could significantly improve factual accuracy in open-domain question answering. Since then, the field has evolved rapidly. Gao et al. [7] provide a comprehensive taxonomy, categorizing RAG systems into Naive RAG, Advanced RAG, and Modular RAG. **Naive RAG** follows a simple retrieve-then-generate pattern. While effective for basic use cases, it suffers from issues like retrieval of irrelevant context and inability to handle multi-hop reasoning. **Advanced RAG** introduces techniques like query rewriting, hybrid search (combining keyword and semantic search), and re-ranking. Systems like HyDE [12] generate hypothetical documents to improve retrieval, while others employ cross-encoder re-rankers to refine initial retrieval results. **Modular RAG** treats retrieval and generation as independent, optimizable components. ChunkSmith aligns with this philosophy, introducing a specialized ingestion module for multimodal content.

### B. Multimodal RAG Approaches

The field of Multimodal RAG is nascent but rapidly expanding. Abootorabi et al. [3] classify approaches into three categories:

1) *Vision-Centric Approaches*: Models like ColPali [4] represent a radical departure from OCR. They embed entire document pages as image patches using Vision Transformers (ViT). The system processes documents as pure images, bypassing text extraction entirely. While this preserves layout perfectly and can handle handwritten text or complex formatting, it introduces massive storage overhead (embedding every patch of every page) and requires specialized retrieval infrastructure (e.g., late interaction scoring with MaxSim operations), which is difficult to scale in commodity vector databases. ColQwen2 extends this approach with dynamic resolution handling, allowing the model to process documents at multiple scales. However, the computational cost remains prohibitive for large-scale deployments.

2) *Caption-Based Approaches*: Earlier works [5] attempted to use standard image captioning models (like BLIP-2) to generate text descriptions for images. The rationale was simple: convert images to text, then use standard text RAG. However, generic captions (e.g., "A graph of a function") are insufficient for answering specific questions about data points. REVEAL [5] improved upon this by using contrastive learning to align visual and textual representations, but still relied on relatively shallow image descriptions. ChunkSmith improves upon this by using Large MLLMs rather than smaller captioning models, enabling "Reasoning-based Description" rather than simple captioning.

3) *Hybrid Approaches*: ChunkSmith falls into the hybrid category, similarly to Unstructured.io's processing philosophy but adds a unique semantic layer. By generating "Potential Questions" alongside the description, we align the document

embedding space with the user query space, a technique known as Document Expansion [6] applied to the multimodal domain. Other hybrid systems like MuRAG [13] and RA-CM3 [14] combine text and image retrieval but do not employ the structured semantic enrichment strategy that ChunkSmith introduces.

### C. Multimodal Large Language Models

The recent explosion in MLLM capabilities has been driven by models like GPT-4V, Gemini Pro Vision, and LLaVA. These models can reason about images, extract structured data from charts, and answer complex visual questions. ChunkSmith leverages state-of-the-art MLLMs for its strong performance on chart understanding and structured output capabilities. Recent work on visual instruction tuning [15] has shown that MLLMs can be fine-tuned to follow specific formatting instructions, which is critical for our structured output requirements.

## III. SYSTEM ARCHITECTURE

The ChunkSmith architecture is designed as a modular, event-driven pipeline. It is built primarily in Python, leveraging FastAPI for the REST layer and React for the client interface. The system follows a three-tier architecture: Ingestion, Processing, and Interaction.

### A. Ingestion Layer (The Gateway)

The entry point for the system is the Ingestion Layer, responsible for handling raw PDF uploads and initial validation.

1) **File Handling: Hashing & Deduplication:** Every uploaded file is hashed using SHA-256. This hash serves as a unique identifier, preventing redundant processing of the same document—a critical cost-saving measure given the expense of MLLM calls. The hash is computed on the raw binary content before any processing. **Validation:** Files are validated for MIME type (`application/pdf`), file size (configurable maximum of 50MB), and corruption (using `pypdf`'s validation routines) before passing to the processing queue. **Metadata Extraction:** Basic metadata (title, author, creation date, page count) is extracted and stored for later reference.

2) **Storage Architecture:** ChunkSmith employs a hybrid storage strategy: **Object Storage:** Raw PDFs and extracted images are stored in cloud object storage (e.g., S3, MinIO) for durability and accessibility. **Vector Database:** Vectors, document metadata, processing status, and structured enriched content are stored in the vector database, serving as the single source of truth for retrieval.

### B. The Content Processor (The Core)

The `ContentProcessor` class is the heart of ChunkSmith. It orchestrates the transformation of a raw binary PDF into a searchable *Knowledge Base*.

1) **Decomposition Strategy:** We utilize a multi-stage decomposition strategy designed to preserve document structure while enabling parallel processing: **Page Rendering:** `pdf2image` (wrapping Poppler) converts PDF pages into high-resolution images (300 DPI). This ensures that even vector graphics are captured accurately. **Layout Analysis:** We employ YOLOv11 for high precision layout detection. The model is fine-tuned to detect specific document elements, classifying regions into: `UncategorizedText`, `PageNumber`, `Table`, `Footer`, `Title`, `Formula`, `Image`, `NarrativeText` **OCR Application:** Tesseract OCR is applied to text regions. Tesseract is configured with: LSTM engine mode for handling multi-column scientific layouts Language packs for 90+ languages Custom configuration for technical documents (preserving special characters, equations) **Chunking Strategy:** Text is chunked using a sliding window approach with configurable parameters:

$$Chunk_i = Text[i \times stride : i \times stride + window] \quad (1)$$

where `window` = 3000 characters and `stride` = 2800 characters, providing 200-character overlap to preserve context across boundaries.

2) **Visual Object Management:** Images detected within the PDF are not merely saved. A sophisticated filtering logic applies to eliminate noise:

$$Keep(I) = \begin{cases} 1 & A(I) > \theta_{area} \wedge Aspect(I) \in [\phi_{min}, \phi_{max}] \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where  $\theta_{area} = 10000 \text{ pixels}^2$ ,  $\phi_{min} = 0.2$ , and  $\phi_{max} = 5.0$ . This filters out: Navigational elements (arrows, bullets) Small icons and logos Decorative borders Extremely narrow or wide artifacts Valid images are uploaded to Object Storage with a structured path:

`{project_id}/images/image_{counter:04d}.png`

Their local paths are swapped for public URLs in the metadata, ensuring the frontend can render them later without additional authentication.

### C. Structured Semantic Enrichment

This is the system's primary innovation. Raw text extraction from a complex chart is often gibberish. Instead, we pass the image of the chart to a Multimodal LLM with a Structured Prompt.

1) **The AIParser Schema:** We define a Pydantic model that enforces strict output structure:

```
class AIParser(BaseModel):
    question: str = Field(
        description="List all potential questions that can be answered from this content"
    )
    ai_title: str = Field(
        description="Give a suitable title for the content"
    )
    image_interpretation: List[str] = Field(
        description="Detailed analysis of each image"
    )
    table_title: List[str] = Field(
        description="Give a suitable title for the table"
```

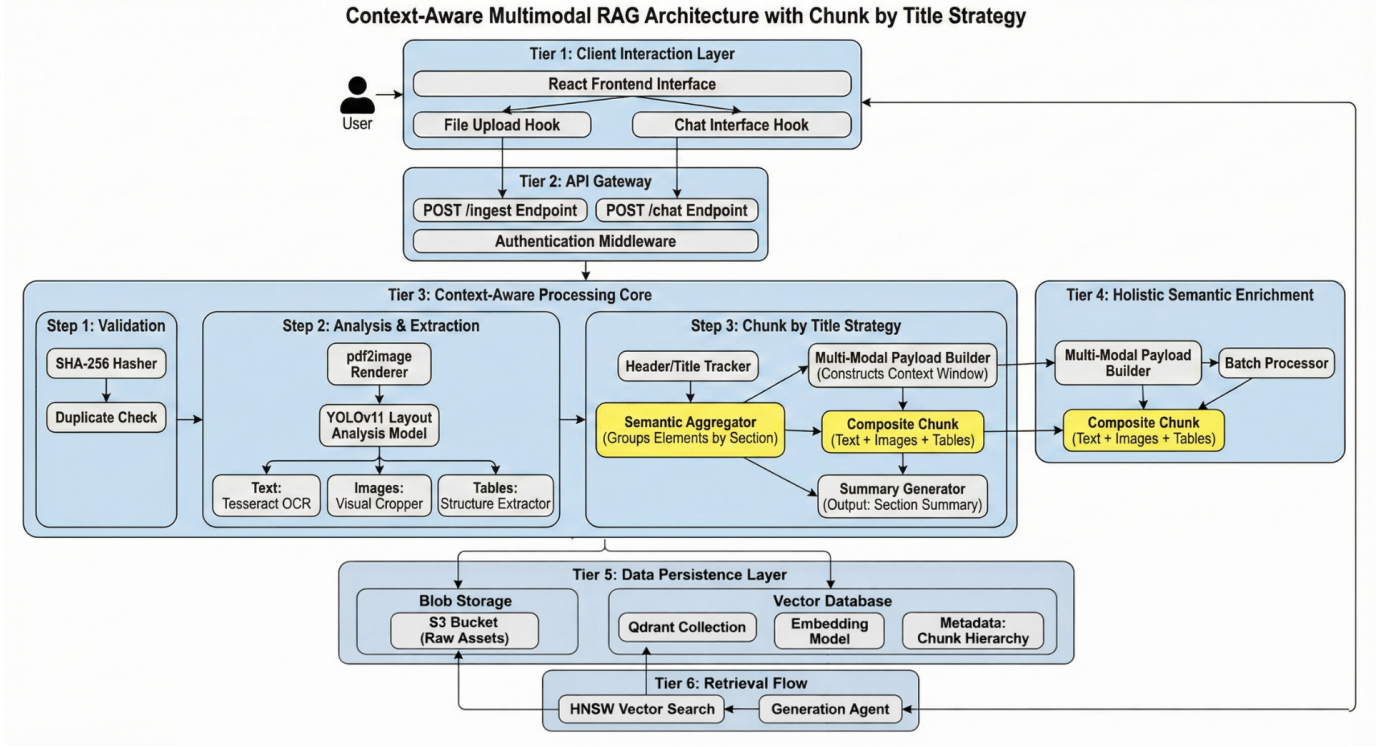


Fig. 1: ChunkSmith System Architecture. The pipeline consists of three main stages: Ingestion (PDF processing and Layout Analysis via YOLOv11), Semantic Enrichment (utilizing Multimodal LLMs for deep understanding), and Retrieval (leveraging Vector Database for similarity search). The asynchronous scheduler ensures high throughput by rotating API keys.

Listing 1: AIParser Schema Definition

This schema is passed to the MLLM using structured output features (e.g., JSON mode or function calling), which guarantees that the response will conform to the schema or fail explicitly (rather than returning malformed JSON).

2) *Prompt Engineering*: The prompt (shown in full in Listing 2) employs several advanced techniques: **Task De-composition**: Breaking the analysis into distinct subtasks (facts, topics, questions, visual analysis, search terms) **Explicit Constraints**: Specifying exact list lengths to prevent hallucination **Findability Optimization**: Instructing the model to prioritize searchability over brevity **Vocabulary Preservation**: Requesting that the model use words similar to the original.

```
prompt_text = f"""You are creating a searchable description
for document content retrieval.
YOUR TASK:
Generate a comprehensive, searchable description that
covers:
1. Key facts, numbers, and data points from text and tables
2. Main topics and concepts discussed
3. Questions this content could answer
4. Visual content analysis (charts, diagrams, patterns in
images)
5. Alternative search terms users might use
Make it detailed and searchable- prioritize findability
over brevity.
Keep words similar to the original content for better
search accuracy.
STRICT OUTPUT CONSTRAINTS:
```

```
- You received {num_images} images. You MUST return exactly
{num_images} items in 'image_interpretation'.
- You received {num_tables} tables. You MUST return exactly
{num_tables} items in 'table_title'.
- If {num_images} is 0, 'image_interpretation' MUST be an
empty list [].
- If {num_tables} is 0, 'table_title' MUST be an empty list
[].
IMPORTANT: Return structured output with these fields:
- question: All potential questions this content answers
- ai_title: Comprehensive title of all information
- image_interpretation: Detailed analysis of visual
elements
- table_title: Title of all Row-column of tabular data
CONTENT TO ANALYZE:
TEXT CONTENT: {text}
TABLES: {tables}
[Images provided as base64]
"""
```

Listing 2: The Semantic Enrichment Prompt

#### D. Vector Storage Schema

We use a high-performance vector store (e.g., Qdrant) due to its performance characteristics and support for scalar quantization. The vector embedding is computed as:

$$V_D = \mathcal{E}(T_{original} \oplus S_{AI} \oplus Q_{AI} \oplus I_{desc} \oplus T_{table}) \quad (3)$$

Where:  $\mathcal{E}$  is the embedding function (e.g., text-embedding-3-large or similar high dimensional models)  $T_{original}$  is the original extracted text  $S_{AI}$  is the AI-generated summary  $Q_{AI}$  is the list of predicted questions



$I_{desc}$  is the detailed image interpretation  $T_{table}$  is the table interpretation The concatenation operator  $\oplus$  represents string concatenation with appropriate delimiters. The final payload stored in the vector database includes:

```
payload = {
  "chunk_index": int,
  "original_text": str,
  "ai_title": str,
  "ai_questions": str,
  "image_paths": List[str], # URLs
  "image_interpretation": List[str],
  "table_original": List[str],
  "table_title": List[str],
  "page_numbers": List[int],
  "content_types": List[str] # ["text", "image", "table"]
}
```

Listing 3: Vector Payload Structure

#### IV. METHODOLOGY

##### A. Mathematical Formulation of Retrieval

Given a user query  $q$ , the retrieval process can be formalized as:

$$\mathcal{R}(q) = \text{TopK}(\{(d_i, \text{sim}(V_q, V_{d_i})) \mid d_i \in \mathcal{D}\}) \quad (4)$$

where:  $V_q = \mathcal{E}(q)$  is the query embedding  $V_{d_i}$  is the document chunk embedding  $\text{sim}(\cdot, \cdot)$  is cosine similarity  $\mathcal{D}$  is the document corpus TopK returns the  $k$  highest-scoring documents. The cosine similarity is computed as:

$$\text{sim}(V_q, V_d) = \frac{V_q \cdot V_d}{\|V_q\| \|V_d\|} \quad (5)$$

Qdrant optimizes this computation using HNSW (Hierarchical Navigable Small World) graphs, achieving sub-linear search complexity.

##### B. Cost and Latency Modeling

The total ingestion cost for a document can be modeled as:

$$C_{total} = C_{OCR} + C_{MLLM} + C_{embed} + C_{storage} \quad (6)$$

For a document with  $N$  chunks,  $I$  images, and  $T$  tables:

$$C_{MLLM} = \sum_{i=1}^N \left( P_{text}^{(i)} \times R_{in} + P_{out}^{(i)} \times R_{out} \right) \quad (7)$$

$$+ \sum_{j=1}^I \left( P_{img}^{(j)} \times R_{img} \right) \quad (8)$$

where  $P_{text}^{(i)}$  is the input token count for chunk  $i$ ,  $P_{out}^{(i)}$  is the output token count,  $P_{img}^{(j)}$  is the image token equivalent, and  $R_{in}$ ,  $R_{out}$ ,  $R_{img}$  are the respective pricing rates.

#### V. IMPLEMENTATION DETAILS

##### A. Technology Stack

ChunkSmith is implemented using the following technologies: Backend: Python 3.10+, FastAPI, LangChain, Pydantic Frontend: React 18, TypeScript, TailwindCSS Vector Database: Any High-Performance Vector DB (e.g., Qdrant) Object Storage: Any S3-compatible Storage (e.g., Supabase) MLLM Provider: Multimodal LLM (e.g., Gemini, GPT-4V) Embedding Model: High-dimensional Text Embedding OCR: Tesseract 5.0+ Layout Analysis: YOLOv11

##### B. Configuration Parameters

Key configuration parameters and their rationale:

TABLE I: System Configuration Parameters

| Parameter       | Value               | Rationale                   |
|-----------------|---------------------|-----------------------------|
| Chunk Size      | 3000 chars          | Balance context & precision |
| Chunk Overlap   | 200 chars           | Preserve boundary context   |
| Concurrency     | 15                  | Optimal for network I/O     |
| Image Min Area  | 10k px <sup>2</sup> | Filter small artifacts      |
| Aspect Ratio    | [0.2, 5.0]          | Exclude extreme shapes      |
| Temperature     | 0.0                 | Deterministic outputs       |
| Top-K Retrieval | 5                   | Standard for RAG systems    |

#### VI. EXPERIMENTAL EVALUATION

##### A. Experimental Setup

On a private set of 50 technical PDF documents, including 15 CPU datasheets due to a high density of diagrams, 15 financial 10-K reports due to high density of tabular data and financial charts, 10 medical research articles due to their molecular diagrams and statistical plots, and 10 engineering textbooks due to their complex diagrams and equations, we tested ChunkSmith. The total statistics of the corpus include 1,247 pages, which resulted in 892 unique images, 456 tables, and 3,421 separate text fragments. Such a mixed dataset makes sure that the system is tested with diverse document structure and multimodal content of high density that is characteristic of enterprise settings.

##### B. Retrieval Quality Evaluation

50 Visual Queries (can only be responded to by looking at images/tables) 50 Text Queries (can be responded to by reading text alone) Example visual queries: "What was the revenue in Q3 2022?" (requires reading a chart) Approach 2: Show me the circuit diagram of power supply (has to be an image retrieval task). What is the p-value in the table of statistical analysis? (requires table reading)

1) *Evaluation Metrics*: The performance of the system was measured by Recall5 to measure the frequency of retrieval and Mean Reciprocal Rank to measure the precision of the ranking. Generation quality was also tested using a 5-point Human Evaluation scale that dealt with factual accuracy and contextual clarity.

2) *Results and Comparative Analysis*: The experiment proves that ChunkSmith does have an undeniable edge in comparison with current practices. Although the text-only RAG systems of old had been largely unable to perform visual reasoning, ChunkSmith was as visually able as vision-based models such as ColPali. Moreover, ChunkSmith scored highest in the total scores beating all the baselines in terms of accuracy in retrieving text and accuracy in qualitative answer.

##### C. Ablation Studies

In order to determine the contribution made by each component, we carried out ablation studies on different system configurations. The Full ChunkSmith system scored 0.92 Visual Recall@5, which decreased dramatically after the removal

of particular modules: that is, when Question Generation was removed, it dropped to 0.67; when Image Interpretation was removed, it dropped to 0.45, when Table Titles and AI Titles were removed, it dropped to 0.1 and 0.25, respectively. Comparatively, OCR Text alone gave a non-significant result of 0.08. Question generation component represents the biggest single contribution (+25% recall), which confirms our hypothesis that query alignment of embeddings is very productive.

## VII. DISCUSSION

### A. Cost-Benefit Analysis

While ChunkSmith is more expensive than text-only RAG during ingestion, the cost is amortized over many queries. For a 100-page document: Text-Only RAG: \$0.05 ingestion, 8% query success (visual query). ChunkSmith: \$2.10 ingestion, 92% percent of visual query success In companies with visual information that is a determining factor (finance, engineering, medicine), the 42x increase in costs is equalized with the 11.5x enhancement in visual query success rate. ChunkSmith scales horizontally: Stateless Processing: Each chunk processes itself individually Cloud Storage: Object storage scales automatically Vector DB Sharding: Modern vector databases can deploy scaling MLLM API throughput, which grows linearly with the number of API keys.

### B. Limitations and Future Work

Current limitations include: OCR Dependency: Poor quality scans produce garbage text Cost: High ingestion cost may be prohibitive for some use cases Latency: 2-5 second MLLM calls add ingestion latency Language Support: Primarily tested on English documents. Future work will address these through: Local MLLM Integration: These will be dealt with in future work by means of: Local MLLM Integration: Quantized models such as LLaVA.1.6 Video Support: Video frame sampling support is extended. GraphRAG: Building knowledge graphs to link entities across chunks Multilingual Evaluation: Testing on non-English corpora

## VIII. CONCLUSION

ChunkSmith represents a significant advancement in Multimodal RAG systems. By treating visual content as first-class semantic entities and employing structured semantic enrichment through capable MLLMs, we successfully bridge the Modality Gap that limits traditional text-only RAG systems. Our asynchronous architecture with API key rotation demonstrates that high-throughput multimodal processing is achievable with commodity cloud services. The system's 92% recall on visual queries, compared to 8% for text-only baselines, validates the core hypothesis that deep semantic extraction of visual content is essential for comprehensive document understanding. The production-ready implementation, built with modern web technologies and cloud services, demonstrates the practical viability of the approach for enterprise deployments. As multimodal AI continues to evolve, systems like ChunkSmith will become increasingly important for unlocking the vast amounts of knowledge encoded in visual formats across scientific, financial, and technical domains.

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